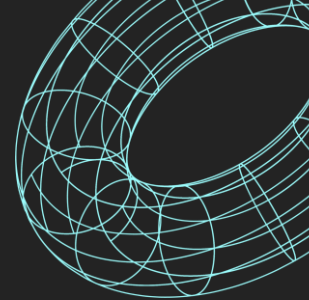




CS290 Project

Tourist Guide System

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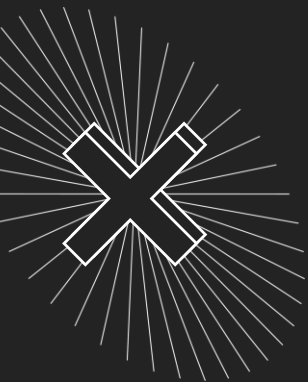


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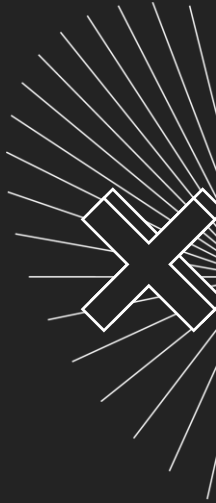
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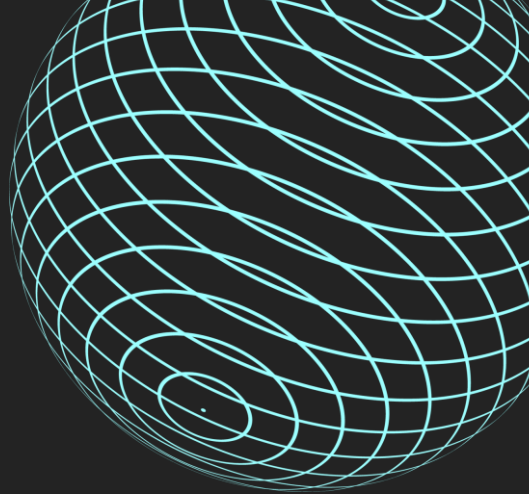
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**Design The
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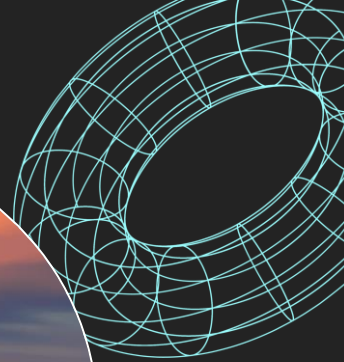
01

Project Initiation



Project Scope

- The Tourist Guide System is developed as a one-stop platform for travelers, offering a range of services including tourist destination insights, real-time transport schedules, and accommodation booking.
- The platform integrates detailed information on tourist attractions, weather, and events, making it easier for users to make real-time decisions and plan their trips effectively.
- It focuses on Saudi Arabia, providing both local and international tourists with an up-to-date travel guide.



Objectives



- **Personalized Travel Plans** Allow users to create custom itineraries tailored to their interests and schedules, ensuring a more personalized travel experience.
- **Real-Time Data** Provide live updates on weather, events, and transport schedules to assist in real-time decision-making.
- **User-Centric Design** Create an intuitive platform that provides users with comprehensive information on tourist attractions, transport options, and events in Saudi Arabia.
- **Data Security** Implement strong data security measures, including encryption and privacy compliance.
- **Cross-Platform Accessibility** Ensure a seamless user experience across devices, with offline functionality for essential features like maps.



Target Audience

Service Providers

Hotels, tour operators, and transportation services that want to connect with tourists through the platform

Tourists

Both international visitors and local tourists who need a reliable tool to explore Saudi Arabia's attractions, book accommodations, and arrange transport.

Adventurous Travelers

People who enjoy exploring unfamiliar places and need an app that provides GPS navigation, real-time updates, and event suggestions.



System Features Vision

Centralized Information Hub

The system will act as the main source for tourist information in Saudi Arabia, ensuring all data is up-to-date and reliable

Interactive Maps & GPS

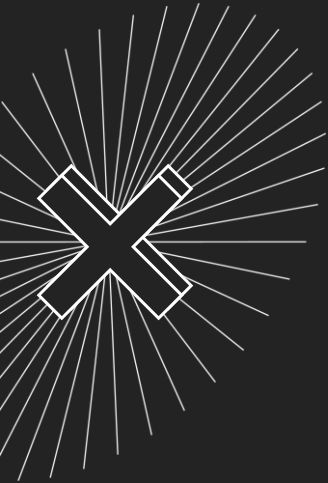
Provide travelers with GPS-enabled maps that help them navigate easily through different destinations

Seamless Booking

Integrate booking features for accommodations and transportation, simplifying the travel planning process

Data Privacy

Implement strict privacy controls, ensuring user data is protected from breaches



Product Backlog Highlights

Destination Information

Detailed insights into tourist spots like historical sites, museums, and natural landmarks in Saudi Arabia

Itinerary Planning

Tools for creating personalized travel plans based on user preferences, schedules, and interests

GPS and Real-time Updates

Integration of live updates for events, weather, and transport schedules, along with GPS-enabled maps



Data Security and Privacy

Strong security measures like encryption and secure login to protect user information

Transportation and Booking

Booking options for transport and accommodations with filters for user preferences like price and location

Multi-Platform Compatibility

Responsive design for seamless performance across multiple devices and operating systems



User and System Requirements



User Requirements

- Access detailed information on popular tourist spots and events
- Create and customize travel itineraries based on personal preferences
- Receive real-time updates on weather, events, and transport schedules
- Have secure access to personal data, itineraries, and preferences

System Requirements

- Provide accurate, real-time data from APIs for weather, transport, and events
- Allow user feedback and offer support services through the website
- Enable offline access to crucial features like maps and travel plans



Sprint Backlog

Tourist Information Interface

Develop user-friendly interfaces for browsing tourist attractions, historical sites, and natural wonders

Real-time Event and Weather Updates

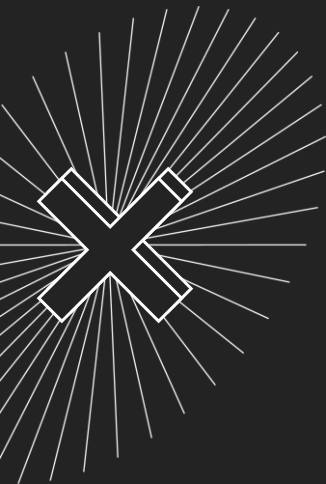
Integrate APIs for real-time information on transport schedules, events, and weather

Personalized Travel Plans

Design an easy-to-use travel itinerary planner, with filters for user preferences

User Data Security

Implement secure authentication methods and test data protection features to ensure compliance with privacy standards



System Model Design

Activity Diagram

Visual representation of user interactions and system activities

Use Case Diagrams

Define various user scenarios, including user login, search functionality, and content filtering

Sequence and Class Diagrams

Illustrate the flow of interactions between system components and the relationships between classes within the system



Architectural Design

MVC (Model-View-Controller) Pattern

- The Model handles data and business logic
- The View manages the presentation layer (UI)
- The Controller processes inputs from the user and communicates between Model and View

Benefits

- The system is modular, making it easier to maintain and scale
- Enhances testing, simplifies feature updates, and promotes a clean code structure
- Separation of concerns ensures smooth handling of data and UI independently

